

Additional Mathematics

Chapter 12: Differentiation

NOTES

12.1 The Derivative and Its Basic Rules

① Notation of Derivative

1. Differentiate with respect to x

$$\frac{d}{dx}(\text{function}) =$$

2. Differentiate y with respect to x

$$\frac{dy}{dx} =$$

3. Differentiate function of x

$$f'(x) =$$

② Power Rule

$$y = x^n \rightarrow \frac{dy}{dx} = nx^{n-1}$$

Bring down power, reduce power by 1.

③ Constant Rule

The derivative of any constant is 0.

④ Constant Multiple Rule

$$y = kx^n \rightarrow \frac{dy}{dx} = knx^{n-1}$$

→ differentiate normally, keeping constant the same

⑤ Sum and difference Rule

$$\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$$

→ differentiate one term at a time.

12.2 The Chain Rule

⑥ Chain Rule

$$\frac{d}{dx}(u^n) = nu^{n-1} \cdot \frac{du}{dx}$$

Differentiate outside to inside.

12.3 The Product Rule

⑦ Product Rule

When there are 2 functions multiplied together,

diff left keep right + keep left diff right

12.4 The Quotient Rule

⑧ Quotient Rule

When there is a function divided by another function,

diff top keep bottom - keep top diff bottom
bottom²

QUESTION TYPES

Simplifying Fractions to Avoid Product/Quotient Rule

Case 1: If bottom is just one term x^n , split the fraction.

Case 2a. If top is a constant:

$$\frac{k}{(\text{function})^n} = k(\text{function})^{-n}$$

Case 2b. If bottom is a constant:

$$\frac{(\text{function})^n}{k} = \frac{1}{k}(\text{function})^n$$

Case 3: Expand if possible (product of two polynomials).

Other Question Types

1. Find gradient of curve at a point

Step 1: Find $\frac{dy}{dx}$

Step 2: substitute x -coordinate into $\frac{dy}{dx}$ to find gradient.

2. Find coordinates of point(s) at which gradient of curve = k

Step 1: Find $\frac{dy}{dx}$.

Step 2: Set $\frac{dy}{dx}$ to k to solve for x .

Step 3: substitute value(s) of x into curve to find y -coordinate, hence finding the coordinate(s) of point(s).

3. Find coordinates of point on curve where gradient = 0.

Step 1: Find $\frac{dy}{dx}$.

Step 2: Set $\frac{dy}{dx} = 0$ and solve for x .

Step 3: Substitute value(s) of x to equation of curve to find y -coordinate.

4. Given gradient of curve at a given point, find unknowns in equation.

Step 1: Find $\frac{dy}{dx}$.

Step 2: Set $\frac{dy}{dx} = \text{gradient}$ and substitute x -coordinate of point (equation 1)

Step 3: Substitute coordinates of point into equation of curve (equation 2)

Step 4: Simultaneous equations to find unknowns.